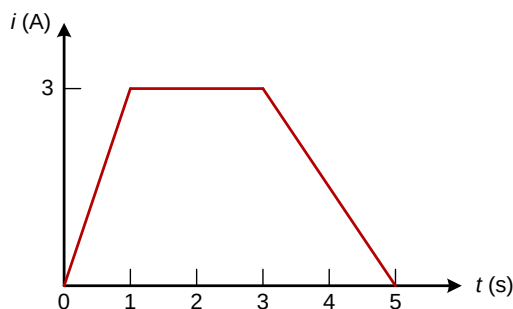


Topic 1: Circuit Basics

1. Determine the total charge flowing through an element for $0 \leq t \leq 2$ seconds when the current entering the positive terminal is $i(t) = e^{-2t}$ mA.

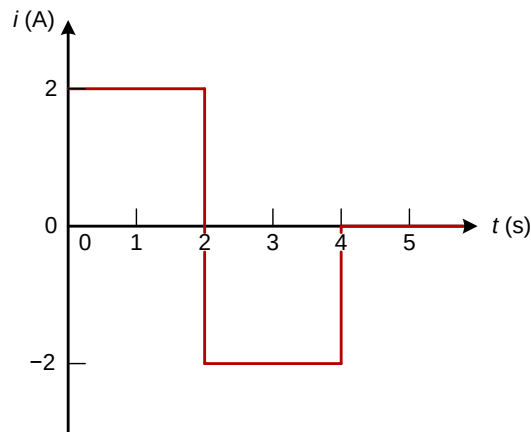
Answer: 0.4908 mC

2. The current flowing through an element is shown in the graph below. Assuming charge entering the element before $t = 0$ is zero, i.e., $q(0) = 0$, calculate the charge that has entered the element at the following times,
 - (a) $t = 1$ s
 - (b) $t = 3$ s
 - (c) $t = 5$ s



Answer: 1.5 C, 7.5 C, 10.5 C

3. If the voltage $v(t)$ across an element is 10 V, and the current through the element $i(t)$ is shown in the following figure, calculate the power and energy and plot their time functions.



Answer:

$$p = vi = 10i = \begin{cases} 20 \text{ W}, & 0 < t < 2 \\ -20 \text{ W}, & 2 \leq t < 4 \\ 0 \text{ W}, & t > 4 \end{cases}$$

$$w(t) = \begin{cases} 20t \text{ J}, & 0 < t < 2 \\ -20t + 80 \text{ J}, & 2 \leq t < 4 \\ 0 \text{ J}, & t > 4 \end{cases}$$

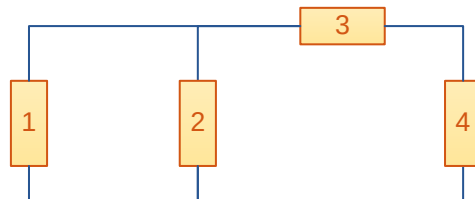
4. How much energy does a 100 W electric bulb consume in one day?

Answer: $2.4 \text{ kWh} = 8.640 \text{ MJ}$

5. The current entering the positive terminal of a device is $i(t) = 6e^{-2t}$ mA and the voltage across the device is $v(t) = 10 di/dt$ V.
- Calculate the power absorbed.
 - Determine the energy absorbed in 3 s.

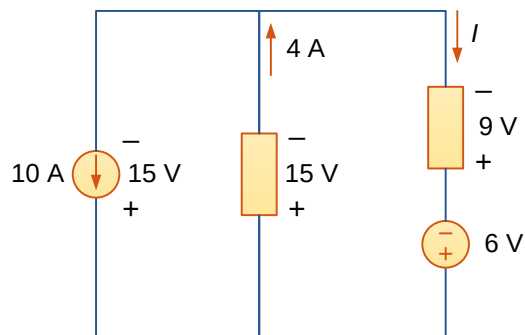
Answer: $-720e^{-4t} \mu\text{W}, -180 \mu\text{J}$

6. The figure below shows a circuit with four elements, $P_1 = 60 \text{ W}$ absorbed, $P_3 = -145 \text{ W}$ absorbed, and $P_4 = 75 \text{ W}$ absorbed.
- How many watts does element 2 absorb?
 - Is element 2 an active element or passive element?



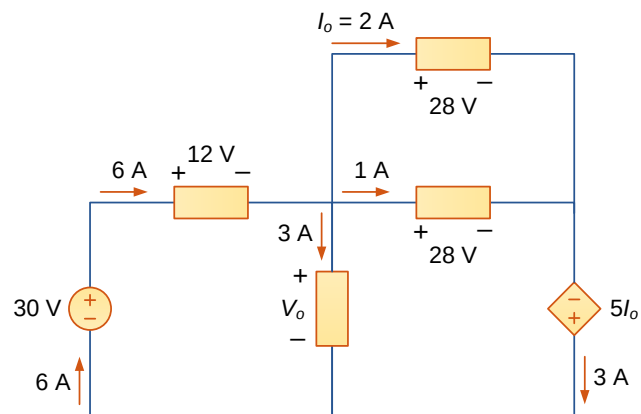
Answer: 10 W absorbed, unknown

7. In the circuit below, find the current I and the power absorbed by each element.



Answer: $-150\text{ W}, 60\text{ W}, 54\text{ W}, 36\text{ W}$

8. In the circuit below, find V_o and the power absorbed by each element.



Answer: $-180\text{ W}, 72\text{ W}, 56\text{ W}, 28\text{ W}, -30\text{ W}, 54\text{ W}$